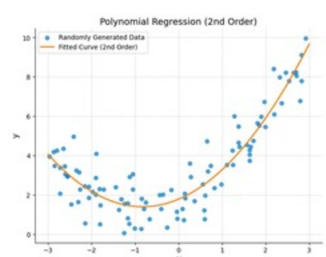
L3 - Regression Techniques in Data Analytics

Regression	A statistical process to estimate the relationships among variables.
Dependent variables	Variables that are being predicted or explained.
Independent variables	Variables that are used to predict the dependent variable.
Linear regression	Best straight line with the lowest error.
Cost function	Quantitatively determines the performance of the model.
Mean absolute error (MAE)	Average of the absolute value of the difference between actual and predicted values.
Mean squared error (MSE)	Average of the squares of the errors, which is the average squared difference between the estimated values and the actual value. $MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$
Root mean squared error (RMSE)	Square root of the mean squared error.
Gradient Descent	A method that initializes the tuning parameters of the regression model randomly and iteratively minimizes the cost.
Learning rate	

	A scalar that controls the pace of going downward in Gradient Descent.
Normal equation	An analytical solution for minimizing the MSE, but not always applicable due to computational constraints.
Singular value decomposition (SVD)	A method that addresses the shortcomings of the Normal equation for minimizing MSE.
Computational cost of Normal equation	Varies from ~ O(n^2.4) to O(n^3) depending on the number of variables.
Computational cost of SVD	Approximately O(n^2), making it more efficient than the Normal equation.
Convex cost function	The cost function (MSE) of linear regression is convex, unlike many other regression models.
Cost of Gradient Descent	Approximately ~ O(m.n) for each iteration, where m is the number of tuning parameters and n is the number of variables.
Batch Gradient Descent	A variant of Gradient Descent that uses the entire dataset to compute the gradient.
Stochastic Gradient Descent	A variant of Gradient Descent that uses a single observation to compute the gradient.
Mini-batch Gradient Descent	A variant of Gradient Descent that uses a small batch of observations to compute the gradient.
Regularized regression	Includes techniques like Ridge regression, Lasso regression, and Elastic Net regression to prevent overfitting.
K-nearest Neighbors regressor	A non-parametric method used for regression that predicts the output based on the k nearest training examples.
K-fold cross-validation	

A technique for assessing how the results of a statistical analysis will generalize to an independent dataset.	
Optimal scenario in cost function minimization	Achieved when the cost function reaches its minimum value.
Expansion of the gradient term	Involves calculating the partial derivatives of the cost function with respect to each variable.
Effects of the learning rate on Gradient Descent	The step size in Gradient Descent is influenced by the learning rate, affecting convergence speed.
Gradient term direction	The gradient shows the upward direction, and its negative is used to minimize the cost.
Tuning parameters	Parameters in a regression model that are adjusted to minimize the cost function.
Stochastic Gradient Descent	It uses a random set of observations (training set) to minimize the cost function at every step.
Gradient Descent	It behaves stochastically (less regular) compared to the Gradient Descent.
Cost function	It is shown using contours.
Main features of Stochastic Gradient Descent	Its results are good (not necessarily optimal).
Stochastic Gradient Descent	It is likely to skip the local minimum in its search to minimize the cost function.
Data shuffling	Always shuffle the data, especially when using Stochastic Gradient Descent because your solution is based on stochasticity (randomness).
Batch	The entire set of data.
Mini-batch	A small set of data (for instance, 100 samples from 10000 observations).

Iteration in Stochastic Gradient Descent	Uses one random sample.
Iteration in Batch Gradient Descent	Uses the entire data at each to minimize the cost function.
Iteration in Mini-batch Gradient Descent	Uses a random set.
Epoch	Each epoch corresponds to going through the entire data once (full pass).
Iteration	Each iteration corresponds to a single update of the model weights (single update).
Polynomial regression	We first define new features based on the original features.
Applying polynomial regression of order 3	New model: y= ¸1 + ¸2 + ¸3.
Polynomial regression of order 2	What if you have two variables and like to apply the polynomial regression of order 2? 
Transform input features for polynomial regression	Linear regression based on other parameters.
Using other variables in linear regression	Is it possible to use other variables in linear regression instead of polynomials?
	We usually divide the data (100%) into training (~80%) and test (~20%) sets.

Data division for training and testing the model	
Tuning parameters	We tune the parameters by minimizing the loss using the training data.
Model performance check	We then check the model performance using the test data.
Validation data	Validation is used to check the model performance during the training.
Velocity	A measure of the speed of an object in a specified direction.
Density	The mass per unit volume of a substance.
Poisson ratio	A measure of the proportional decrease in diameter to the proportional increase in length when a material is stretched.
Resistivity	A measure of how strongly a material opposes the flow of electric current.
Gamma Ray	High-energy electromagnetic radiation emitted by radioactive decay.
Sandstone	A type of sedimentary rock composed mainly of sand-sized mineral particles.
Shale	A fine-grained sedimentary rock formed from mud that is a mix of flakes of clay minerals and tiny fragments of other minerals.
Siltstone	A sedimentary rock composed mainly of silt-sized particles.
Underfitting	A scenario where the model is not complex enough for the problem, resulting in

	high cost function values for training and test data.
Overfitting	A scenario where the model is too complex for the problem, performing well on training data but poorly on test data.
Cost function	A measure used to evaluate how well a model performs, with lower values indicating better performance.
RMSE	Root Mean Square Error, a common measure of the differences between predicted and observed values.
MSE	Mean Squared Error, the average of the squares of the errors between predicted and observed values.
Bias	The error introduced by approximating a real-world problem, which can lead to underfitting.
Variance	The error introduced by the model's sensitivity to fluctuations in the training set, which can lead to overfitting.
Regularization	A technique used to prevent overfitting by constraining the weights of the model.
Ridge regression	A type of linear regression that uses L2 norm squared for regularization.
	A type of linear regression that uses L1 norm for regularization, setting less-important weights to zero.

Lasso regression	
Elastic Net regression	A regression method that combines both Ridge and Lasso regression techniques.
Hyperparameter	A parameter whose value is set before the learning process begins, influencing the training process.
K-nearest Neighbors Regressor	A regression model that predicts the output based on the closest training examples in the feature space.
K-fold Cross-validation	A technique for assessing how the results of a statistical analysis will generalize to an independent data set.
Early stopping	A technique used to prevent overfitting by stopping training when the validation loss reaches a minimum.
Training data	The dataset used to train a model.
Validation data	The dataset used to evaluate the model during training to prevent overfitting.
Epoch	One complete pass through the entire training dataset.
Loss	A measure of how well the model's predictions match the actual data.

Feature selection	The process of selecting a subset of relevant features for model construction.
Ridge regression	Usually better than the original regression because regularizations improve performance.
Lasso and Elastic Net regressions	Better than Ridge regression if you plan to check the effects of the number of tuning variables.
Lasso regression	Allows you to transition your model from Ridge to Lasso regression.
Early stopping	Helps avoid overfitting by stopping training when validation loss reaches a minimum.
K-nearest Neighbors Regressor	A regression model that does not require defining new features and can handle oscillation and nonlinearity.
K-fold Cross-validation	A technique used to evaluate the performance of a model by dividing the data into K subsets.
Logistic regression	Determines the probability of a sample being part of a specific group, such as the probability of a hydrocarbon reservoir being economically producible.
Logistic function	A function used in logistic regression to provide probabilities, defined as 1/(1+e^(-z)).
Derivative of the logistic function	The derivative of the sigmoid function, which is used in optimization during logistic regression.
Cost function of Logistic regression	A function used to measure the performance of a logistic regression model, along with its partial derivative.
Confusion matrix	

A table used to evaluate the performance of a classification model by comparing predicted and actual values.

Number of neighbors in KNN	Changing the number of neighbors affects the model performance in K-nearest Neighbors regression.
Best number of neighbors	The optimal number of neighbors that yields the best performance in K-nearest Neighbors regression.
Interpolation in KNN	KNN is more powerful than Linear Regression for interpolation tasks.
Extrapolation in KNN	KNN usually fails to extrapolate beyond the training data.
Overfitting	Occurs when a model learns the training data too well, failing to generalize to new data.
Analytics Question	Can we create a classification logistic regression model to predict whether an employee is quitting or not using the HR data set?